

# Global Illumination for Fun and Profit

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Figure 1: Dramatic evening clouds.

## ABSTRACT

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**Index Terms:** Radiosity, global illumination, constant time.

## 1 INTRODUCTION

This template is for papers of VGTC-sponsored conferences which are *not* published in a special issue of TVCG.

## 2 USING THIS TEMPLATE

- If you receive compilation errors along the lines of “Package ifpdf Error: Name clash, \ifpdf is already defined” then please add a new line “\let\ifpdf\relax” right after the “\documentclass[journal]{vgtc}” call. Note that your error is due to packages you use that define “\ifpdf” which is obsolete (the result is that \ifpdf is defined twice); these packages should be changed to use ifpdf package instead.

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- The style uses the hyperref package, thus turns references into internal links. We thus recommend using “\cref{reference}” (instead of “Figure~\ref{reference}” or similar) since “\cref{reference}” turns the entire reference into an internal link, not just the number. Examples: Fig. 2 and Tab. 1.
- The style automatically looks for image files with the correct extension (eps for regular L<sup>A</sup>T<sub>E</sub>X; pdf, png, and jpg for pdfL<sup>A</sup>T<sub>E</sub>X), in a set of given subfolders (figures/, pictures/, images/). It is thus sufficient to use “\includegraphics{clouds}” (instead of “\includegraphics{figs/clouds.jpg}”).
- For adding hyperlinks and DOIs to the list of references, you can use “\bibliographystyle{abbrv-doi-hyperref-narrow}” (instead of “\bibliographystyle{abbrv}”). It uses the doi and url fields in a bibT<sub>E</sub>X entry and turns the entire reference into a link, giving priority to the doi. The doi can be entered with or without the “http://dx.doi.org/” url part. See the examples in the bibT<sub>E</sub>X file and the bibliography at the end of this template.

**Note 1:** occasionally (for some L<sup>A</sup>T<sub>E</sub>X distributions) this hyper-linked bibT<sub>E</sub>X style may lead to **compilation errors** (“pdfendlink ended up in different nesting level ...”) if a reference entry is broken across two pages (due to a bug in hyperref). In this case make sure you have the latest version of the hyperref package (i.e., update your L<sup>A</sup>T<sub>E</sub>X installation/packages) or, alternatively, revert back to “\bibliographystyle{abbrv-doi-narrow}” (at the expense of removing hyperlinks from the bibliography) and try “\bibliographystyle{abbrv-doi-hyperref-narrow}” again after some more editing.

**Note 2:** the “-narrow” versions of the bibliography style use the font “PTSansNarrow-TLF” for typesetting the DOIs

in a compact way. This font needs to be available on your L<sup>A</sup>T<sub>E</sub>X system. It is part of the “paratype” package, and many distributions (such as MikTeX) have it automatically installed. If you do not have this package yet and want to use a “-narrow” bibliography style then use your L<sup>A</sup>T<sub>E</sub>X system’s package installer to add it. If this is not possible you can also revert to the respective bibliography styles without the “-narrow” in the file name.

DVI-based processes to compile the template apparently cannot handle the different font so, by default, the template file uses the `abbrv-doi` bibliography style but the compiled PDF shows you the effect of the `abbrv-doi-hyperref-narrow` style.

### 3 BIBLIOGRAPHY INSTRUCTIONS

- Sort all bibliographic entries alphabetically but the last name of the first author. This L<sup>A</sup>T<sub>E</sub>X/bibT<sub>E</sub>X template takes care of this sorting automatically.
- Merge multiple references into one; e.g., use [3, 5] (not [3][5]). Within each set of multiple references, the references should be sorted in ascending order. This L<sup>A</sup>T<sub>E</sub>X/bibT<sub>E</sub>X template takes care of both the merging and the sorting automatically.
- Verify all data obtained from digital libraries, even ACM’s DL and IEEE Xplore etc. are sometimes wrong or incomplete.
- Do not trust bibliographic data from other services such as Mendeley.com, Google Scholar, or similar; these are even more likely to be incorrect or incomplete.
- Articles in journal—items to include:
  - author names
  - title
  - journal name
  - year
  - volume
  - number
  - month of publication as variable name (i.e., {jan} for January, etc.; month ranges using {jan #}/{# feb} or {jan #}{--}{# feb})
- use journal names in proper style: correct: “IEEE Transactions on Visualization and Computer Graphics”, incorrect: “Visualization and Computer Graphics, IEEE Transactions on”
- Papers in proceedings—items to include:
  - author names
  - title
  - abbreviated proceedings name: e.g., “Proc.\ CONF.ACRONYM” without the year; example: “Proc.\ CHI”, “Proc.\ 3DUI”, “Proc.\ Eurographics”, “Proc.\ EuroVis”
  - year
  - publisher
  - town with country of publisher (the town can be abbreviated for well-known towns such as New York or Berlin)

- article/paper title convention: refrain from using curly brackets, except for acronyms/proper names/words following dashes/question marks etc.; example:

- paper “Marching Cubes: A High Resolution 3D Surface Construction Algorithm”
- should be entered as “{M}arching {C}ubes: A High Resolution {3D} Surface Construction Algorithm” or “{M}arching {C}ubes: A high resolution {3D} surface construction algorithm”
- will be typeset as “Marching Cubes: A high resolution 3D surface construction algorithm”

- for all entries
  - DOI can be entered in the DOI field as plain DOI number or as DOI url; alternative: a url in the URL field
  - provide full page ranges AA--BB
- when citing references, do not use the reference as a sentence object; e.g., wrong: “In [4] the authors describe ...”, correct: “Lorensen and Cline [4] describe ...”

### 4 SUPPLEMENTAL MATERIAL INSTRUCTIONS

In support of transparent research practices and long-term open science goals, you are encouraged to make your supplemental materials available on a publicly-accessible repository. Please describe the available supplemental materials in the [Supplemental Materials](#) section. These details could include (1) what materials are available, (2) where they are hosted, and (3) any necessary omissions.

### 5 FIGURE CREDITS

In the [Figure Credits](#) section at the end of the paper, you should credit the original sources of any figures that were reproduced or modified. Include any license details necessary, as well as links to the original materials whenever possible. For credits to figures from academic papers, include a citation that is listed in the **References** section. An example is provided [below](#).

### 6 EXAMPLE SECTION

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### 7 EXPOSITION

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$$\sum_{j=1}^z j = \frac{z(z+1)}{2} \quad (1)$$

Table 1: VIS/VisWeek accepted/presented papers: 1990–2016.

| year | Vis/SciVis | SciVis conf | InfoVis | VAST | VAST conf | TVCG @ VIS | CG&A @ VIS | VIS/VisWeek incl. TVCG/CG&A | VIS/VisWeek w/o TVCG/CG&A |
|------|------------|-------------|---------|------|-----------|------------|------------|-----------------------------|---------------------------|
| 2016 | 30         |             | 37      | 33   | 15        | 23         | 10         | 148                         | 115                       |
| 2015 | 33         | 9           | 38      | 33   | 14        | 17         | 15         | 159                         | 127                       |
| 2014 | 34         |             | 45      | 33   | 21        | 20         |            | 153                         | 133                       |
| 2013 | 31         |             | 38      | 32   |           | 20         |            | 121                         | 101                       |
| 2012 | 42         |             | 44      | 30   |           | 23         |            | 139                         | 116                       |
| 2011 | 49         |             | 44      | 26   |           | 20         |            | 139                         | 119                       |
| 2010 | 48         |             | 35      | 26   |           |            |            | 109                         | 109                       |
| 2009 | 54         |             | 37      | 26   |           |            |            | 117                         | 117                       |
| 2008 | 50         |             | 28      | 21   |           |            |            | 99                          | 99                        |
| 2007 | 56         |             | 27      | 24   |           |            |            | 107                         | 107                       |
| 2006 | 63         |             | 24      | 26   |           |            |            | 113                         | 113                       |
| 2005 | 88         |             | 31      |      |           |            |            | 119                         | 119                       |
| 2004 | 70         |             | 27      |      |           |            |            | 97                          | 97                        |
| 2003 | 74         |             | 29      |      |           |            |            | 103                         | 103                       |
| 2002 | 78         |             | 23      |      |           |            |            | 101                         | 101                       |
| 2001 | 74         |             | 22      |      |           |            |            | 96                          | 96                        |
| 2000 | 73         |             | 20      |      |           |            |            | 93                          | 93                        |
| 1999 | 69         |             | 19      |      |           |            |            | 88                          | 88                        |
| 1998 | 72         |             | 18      |      |           |            |            | 90                          | 90                        |
| 1997 | 72         |             | 16      |      |           |            |            | 88                          | 88                        |
| 1996 | 65         |             | 12      |      |           |            |            | 77                          | 77                        |
| 1995 | 56         |             | 18      |      |           |            |            | 74                          | 74                        |
| 1994 | 53         |             |         |      |           |            |            | 53                          | 53                        |
| 1993 | 55         |             |         |      |           |            |            | 55                          | 55                        |
| 1992 | 53         |             |         |      |           |            |            | 53                          | 53                        |
| 1991 | 50         |             |         |      |           |            |            | 50                          | 50                        |
| 1990 | 53         |             |         |      |           |            |            | 53                          | 53                        |
| sum  | 1545       | 9           | 632     | 310  | 50        | 123        | 25         | 2694                        | 2546                      |

## 7.1 Lorem ipsum

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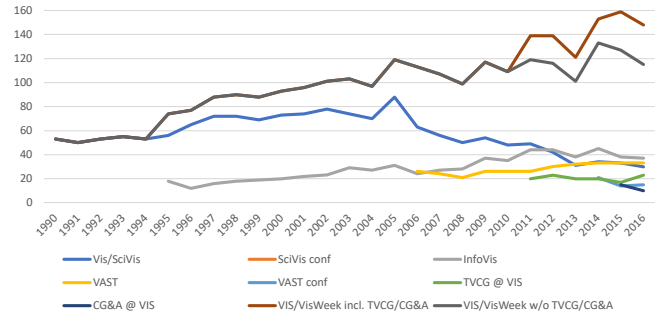


Figure 2: A visualization of the 1990–2016 data from Tab. 1, recreated based on Fig. 1 from [1] and is in the public domain.

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## 8 CONCLUSION

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<sup>1</sup>The algorithm behind Marching Cubes [4] had already been described by Wyvill et al. [7] a year earlier.

<sup>2</sup>Footnotes appear at the bottom of the column.

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## SUPPLEMENTAL MATERIALS

Refer to the instructions for this section (Sec. 4). Below is an example you can follow that includes the actual supplemental material for this template:

All supplemental materials are available on OSF at <https://doi.org/10.17605/OSF.IO/2NBSG>, released under a CC BY 4.0 license. In particular, they include (1) Excel files containing the data for and analyses for creating Tab. 1 and Fig. 2, (2) figure images in multiple formats, and (3) a full version of this paper with all appendices. Our other code is intellectual property of a corporation—Starbucks Research—and there is no feasible way to share it publicly.

## FIGURE CREDITS

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Figure 1 image credit: Dominik Moritz, licensed under CC0.

Figure 2 is a partial recreation of Fig. 1 from [1], which is in the public domain.

## ACKNOWLEDGMENTS

The authors wish to thank A, B, and C. This work was supported in part by a grant from XYZ.

## REFERENCES

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